MIL-S-14642A
12 May 1980
Superseding
MIL-S-0014642 (IH)
15 November 1974

MILITARY SPECIFICATION

SHOULDER BOARD AFROTC CADETS

1. SCOPE.

1.1 Scope--This specification covers one type of shoulder board in two styles for use by AFROTC cadets.

1.2 Classification--Shoulder boards covered by this specification shall be of the following styles (see 6.2):

Style 1—Male

Style 2—Female

1.2.1 Ribbon--The size of the ribbon shall be 1/2 inch for both styles (see 6.2 or 6.3).

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-F-206--Felt Sheet: Cloth, Felt, Wool, Pressed

UU-P-268--Paper, Kraft, Wrapping

V-T-276--Thread, Cotton

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PPP-B-676--Boxes, Setup

PPP-B-636--Box, Shipping Fiberboard

PPP-T-60 --Tape, Packaging, Waterproof

MILITARY

MIL-B-3461--Buttons, Insignia, Metal, Uniform and Cap

MIL-S-3951--Service Ribbons and Bars, Service Ribbon

STANDARDS

MILITARY

MIL-STD-105--Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129--Marking for Shipment and Storage

DRAWING

THE INSTITUTE OF HERALDRY

B-10-11--Shoulder Boards, AFROTC

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Figure 2 is a miniature reproduction of drawing B-10-11 and is attached for information purposes.)

2.2 Other publications--The following standard forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply:

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

US Army Color Card, The Official Colors for Branches, Agencies, etc.

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(Application for copies of the color card should be addressed to The Color Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

3. REQUIREMENTS.

3.1 First Article--Unless otherwise specified (see 6.2) before production is commenced, a sample of the finished shoulder board shall be submitted, or made available, to the contracting officer or his authorized representative for inspection and testing as specified in 4.2. The approval of the first article authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification. The first article shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Material--Materials shall conform to the referenced specifications and the requirements specified herein.

3.2.1 Wool felt--The wool felt to be used to cover the outside of the shoulder board shall conform to type III, Classification No. 6A1 of C-F-206 and shall show good fastness to wet-dry cleaning, crocking, light and water. The felts shall be dyed black to match Cable No. 65018.

3.2.2 Backing cloth--The backing cloth shall be a light weight rayon satin cloth dyed black to match cable No. 65018 and shall show not less than good fastness to crocking, wet-dry cleaning, light and water.

3.2.3 Cement--The cement used for cementing the felt to the spine shall be any type of commercial cement suitable for the purpose.

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3.2.4 Thread--Thread for hand stitching the backing cloth to the felt shall be black cotton thread conforming to type IB3 of V-T-276.

3.2.5 Ribbon bar--The ribbon bar used as an attaching device shall be a double bar conforming to size 2 of MIL-S-3951. The ribbon bar shall be furnished with clutches.

3.2.6 Button--The button mounted on the shoulder board shall be the standard 25 ligne Air Force button conforming to MIL-B-3461.

3.2.7 Paperboard--Paperboard used to line the satin shall be any color, 2 ply bristol board, 1/32 inch in thickness.

3.2.8 Spine--The spine of the shoulder board may be made of either fiberboard or aluminum. When fiberboard is used it shall be a vulcanized fiber 0.060 ($\pm$ 0.005) inch in thichness and when aluminum is used, it may be any grade commercial aluminum, soft temper, a minimum of 1/32 inch thick.

3.2.9 Ribbon--The ribbon for the shoulder board shall conform to the physical requirements specified in table 1 and pattern layout and weave as illustrated in Figure 1.

TABLE 1

Physical Requirements

Width of Ribbon Inches (+1/16-0)	Total Warp Ends (Min)		Picks Per Inch (Min)	Thickness in Inches (Min)	Wt Per Linear Yd (Min)
	Regular	Binder			
1/2	96	4	200	.040	.220 oz

3.2.9.1 Yarn--

3.2.9.1.1 Warp--The regular warp yarn shall be 50/3 ply white cotton. The binder warp yarn shall be 40/1 denier bright nylon, 10 turns per inch.

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3.2.9.1.2 Filling--The filling shall be a silver color mylar plied with a nylon supporting yarn and shall be equivalent to number 12 Laminette. The filling yarn shall be approximately 420 denier, .0085 inches in diameter and round.

3.3 Design--The design of the shoulder board shall be in accordance with Drawing B-10-11.

3.4 Construction--The spine of the shoulder board (see 3.2.8) shall be formed and the front covered with felt specified in 3.2.1. The felt shall extend onto the back, at least 3/4 inch around. The felt shall be glued to the spine with cement specified in 3.2.3. The ribbon bar specified in 3.2.5 shall be centered on and cemented to the bottom of the spine. The back of the shoulder board shall be covered with backing cloth (see 3.2.2) lined with paperboard specified in 3.2.7. The face and 1/2 inch of the back edges of the paperboard shall be covered with backing cloth specified in 3.2.2. The backing cloth shall be cemented to the paperboard and stitched to the felt covering. The button specified in 3.2.6 shall be mounted in the position shown. The button may be secured to the shoulder board by riveting or any method suitable to the contracting officer (see 6.2).

3.4.1 Stitching--The backing cloth shall be secured to the felt by hand stitching using thread specified in 3.2.4 with a minimum of 8 stitches to the inch.

3.5 Workmanship--Shoulder board shall be clean, well made, and shall meet the acceptable quality levels established by this specification.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection--Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection--When first articles are required, inspection and testing of the first article shall be made on a completely fabricated shoulder board for all provisions of this specification applicable to the end product examination and tests.

4.3 Inspection for acceptance--Inspection shall be in accordance with MIL-STD-105 except where otherwise indicated.

4.3.1 Testing of components--In addition to the quality assurance provisions of subsidiary specifications and drawings, components shall be tested as specified in table II. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table II, they shall be as required by the referenced test method. All requirements are applicable for the individual sample unit. The inspection level shall be S-1 and the acceptable quality level for sample unit requirements shall be 6.5 defects per hundred units. The sample unit for the tests in table II shall be as follows:

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Component	Sample Unit
Wood Felt	1 Yard
Backing Cloth	1 Yard
Ribbon	1 Yard
Aluminum or fiberboard	4 Inch square

TABLE II - Component Testing

Component	Characteristic	Reqm't Para	Test Method	# Determ per unit of Product	Results Reported as
Wool Felt for cover	Colorfastness to:				
	Wet-Dry Cleaning	3.2.1	5622	-	Pass or Fail
	Crocking	3.2.1	5651	-	Pass or Fail
	Light	3.2.1	5660	-	Pass or Fail
	Water	3.2.1	5630	-	Pass or Fail
Backing Cloth	Colorfastness to:				
	Wet-Dry Cleaning	3.2.2	5622	-	Pass or Fail
	Crocking	3.2.2	5651	-	Pass or Fail
	Light	3.2.2	5660	-	Pass or Fail
	Water	3.2.2	5630	-	Pass or Fail
Ribbon	Material Indent.	3.2.9.1.1 and 3.2.9.1.2	Std Com'l	1	Pass or Fail
	Picks per inch	Table I	Visual	1	Pass or Fail
	Thickness	Table I	Visual	1	Pass or Fail
	Weight	Table I	5041	-	Pass or Fail
	Weave	3.2.9	Visual	1	Pass or Fail
Aluminum	Material Indent.	3.2.8	Std Com'l	1	Pass or Fail
	Thickness	3.2.8	Std Com'l	1	Pass or Fail
Fiberboard	Material Indent.	3.2.8	Std Com'l	1/ 1	Pass or Fail
	Thickness	3.2.8	Std Com'l	1/ 1	Pass or Fail

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be accepted for the stated requirements.

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4.3.2 End item examination—

4.3.2.1 Visual examination of shoulder board--Visual examination found during this examination shall be classified in accordance with the defects listed herein. The sample unit for this examination shall be one finished shoulder board. The acceptable quality level (AQL) shall be 6.5 defects per 100 units. The inspection level shall be II.

Examine	Defects
Material:	Not as specified.
Felt and Backing Cloth	Shade bars affecting appearance.
	Hole, float, or slub which affects appearance.
	Felt poorly constructed not firm or frayed.
	Not correct color.
Metal	Not Aluminum.
Fiberboard	Not as specified.
Construction	Edges of felt not turned under a minimum of 3/4 inch.
	Edges of backing cloth not turned over paperboard a a minimum of 1/2 inch.
	Edges of felt not cemented to spine.
	Edges of backing cloth not cemented to paperboard.
	Ribbon bar not securely cemented to spine.
	Clutches missing.
	Button not firmly secured to shoulder board.
	Backing cloth not stitched to felt with at least eight stitches per inch.
	Shoulder board stained or spotted.
	Shoulder board not well made, appears shoddy.

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4.3.2.2 Defects in dimensions--Any dimension not within the specified tolerance shall be classified a defect. The acceptable quality level shall be 2.5 defects per hundred units and the inspection level shall be S-3. The sample unit shall be one finished shoulder board.

4.3.2.3 Visual examination of ribbon--Defects found during the examination shall be classified in accordance with the lists shown in 4.3.2.4, 4.3.2.5, and 4.3.2.6.

4.3.2.4 Yard-by-yard examination--The required yardage of each roll shall be inspected and visual defects classified as listed. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be one linear yard. The sample size shall be based upon inspection level II. The lot size shall be expressed in units of one linear yard. The number of rolls from which the sample is to be selected shall be in accordance with Table III. An approximately equal number of yards shall be examined in each roll. The acceptable quality level shall be 4.0 defects per 100 units.

Defects

Hold, cut, or tear.

Broken or missing yarn.

Float or skip.

Open place or crack.

Thick or thin area filling wise.

Incorrect weave in pattern.

Spot, stain, or streak.

Overall width less than specified.

Overall width more than specified.

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4.3.2.5 Overall examination--Each defect listed shall be counted not more than once in each roll examined. The sample unit shall be one roll. The number of rolls selected for yard-by-yard examination shall comprise the sample selected for this examination and the number of defects acceptable shall be as specified in table III.

Defects

Soiled throughout

Finish other than specified.

Selvage curled, folded, rolled, slack or sloppy, torn continuously or intermittently over more than one-half the roll.

TABLE III--ACCEPTABLE NUMBER OF DEFECTS

Lot Size in Yards	Sample Size in Rolls	Maximum Number of Defects Applicable to Sample
Up to 1,300 [1/]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 and up	25	1

[1/] If lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.3.2.6 Examination for Length--

4.3.2.6.1 Individual Rolls--The roll shall be examined for gross length and number and length of pieces on the roll. Any gross length found to be less than the minimum specified or more than 2 yards below the gross length marked on the roll shall be considered a defect with respect to length. The sample unit for this examination shall be one roll. The number of rolls selected as sample (sample size) for this examination and the acceptance number shall be in accordance with Table III.

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4.3.2.6.2 Total yardage in sample--The lot shall be unacceptable if the total of the actual gross lengths of the rolls in the sample is less than the total of the gross lengths marked on the rolls.

4.3.3 Examination of preparation for delivery requirements--An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with and defects shall be scored in accordance with the list below. The samele unit shall be one shipping container fully package, selected just prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, or improper size, location sequence, or method of application.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as, incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
Contents	Number per container is more or less than required.
Weight	Net weight exceeds permissible maximum.

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5. PREPARATION FOR DELIVERY

5.1 Packaging-- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A--

5.1.1.1 Unit packaging--

5.1.1.1.1 Shoulder boards--Two (2) shoulder boards shall be packaged in a snug fitting polyethylene bag and heat sealed. Prior to heat sealing excess air shall be removed.

5.1.1.1.2 Ribbon--Twenty-five yards of ribbon shall be mounted on a spool and wrapped paper with not more than three pieces to a spool. No single piece within the spool shall be less than 5 yards in length. When a spool is made up of more than one piece, the pieces shall be butted and spliced with tape conforming to type III, class 2 of PPP-T-60. Pins or other means of splicing shall not be permitted.

5.1.1.2 Intermediate packaging--Twenty-five unit packages of shoulder boards or five unit packages of ribbon shall be placed in a snug-fitting paperboard box conforming to type 1, style 4 of PPP-B-676. Closure shall be in accordance with the appendix of the specification.

5.1.2 Level C--Shoulder boards and ribbon shall be packaged to afford adequate protection against deterioration and damage during shipment to the first receiving station.

5.2 Packing--Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A--Two hundred and fifty pair of shoulder boards, of one type only, or fifty rolls of ribbon (10 intermediate packages shall be packed in a snug-fitting fiberboard shipping container conforming to class weather-resistant, type V3c or V3s of PPP-B-636. Closure and waterproofing shall be in accordance with the appendix of PPP-B-636.

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5.2.2 Level B--Two hundred and fifty pairs of shoulder boards, of one type only, or fifty rolls of ribbon (10 intermediate packages) shall be packed in a snug-fitting fiberboard shipping container conforming to class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with Method II of Appendix of PPP-B-636.

5.2.3 Level C--Shoulder boards or ribbon packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination.

5.3 Marking--In addition to any special marking required by the contract or purchase order, unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use--Shoulder boards covered by this specification are intended to be used by AFROTC students to indicate rank. The rank will be designated by striping the shoulder board with one or more 1/2 inch metal ribbons.

6.2 Ordering data--Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Style of shoulder board required (see 1.2).

c. When ribbon is required (see 1.2.1).

d. When a first article is not required (see 3.1).

e. When riveting the button to the shoulder board is the only method allowed (see 3.4).

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f. Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.3 Combination of metallic ribbon shall be anchored to the shoulder board at the unit level to distinguish individual grade, but should be procured as a separate item for issue.

Custodians:

Air Force--11

Review Activities:

Air Force--99

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Preparing Activity:

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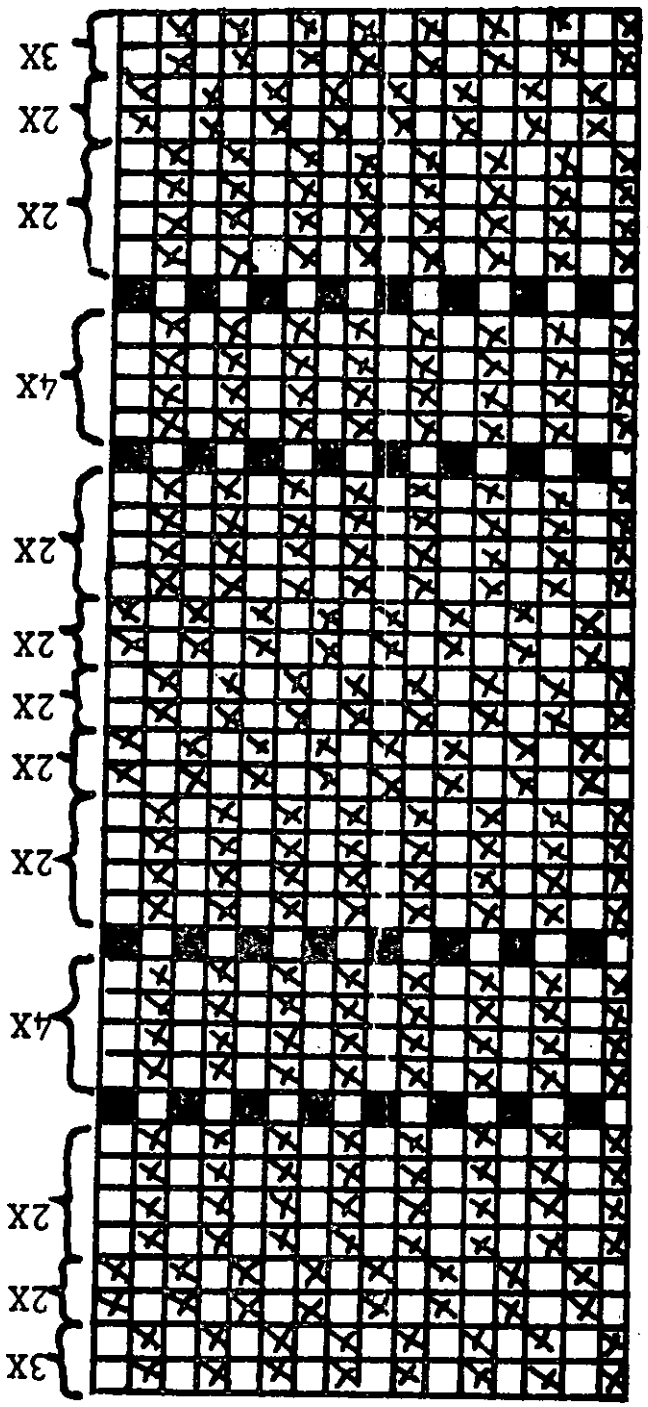
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PATTERN LAYOUT FOR 1/2 IN RIBBON



TOTAL WARP ENDS - 100

X Regular Warp Ends

Binder Warp Ends

FIGURE 1

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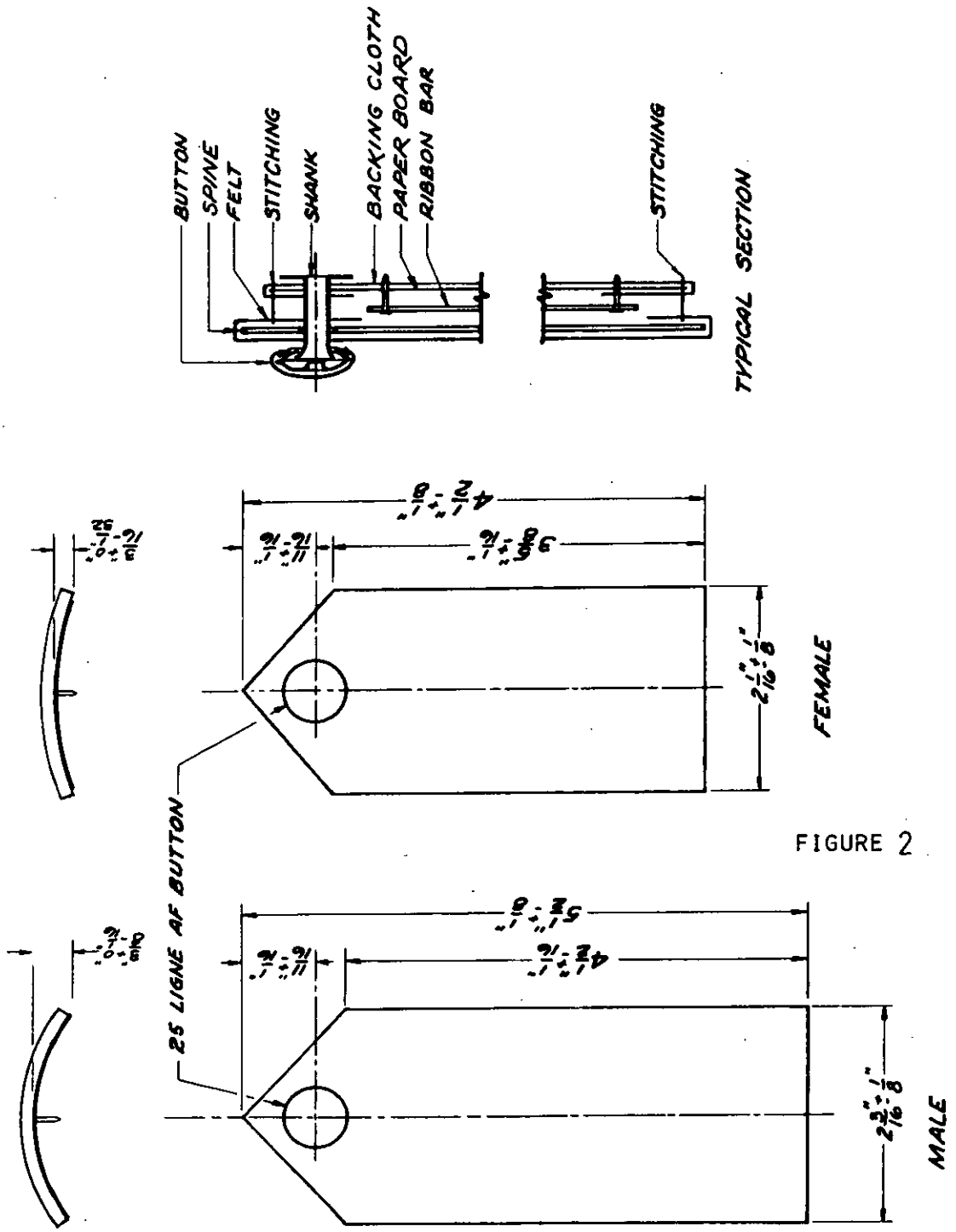


FIGURE 2

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-S-14642A Shoulder Board AFROTC Cadets

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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